Railway Fare and Reservation Calculator

1. Problem Statement

Calculate fares for multiple passengers using distance slabs, travel classes, age-based concessions, reservation fees, and specified cancellation penalties.

2. Objectives

1. Read the number of passengers.
2. Calculate fares based on distance slabs.
3. Apply travel class charges.
4. Provide age-based concessions.
5. Add reservation fees.
6. Calculate cancellation penalties.
7. Display the total fare for all passengers.

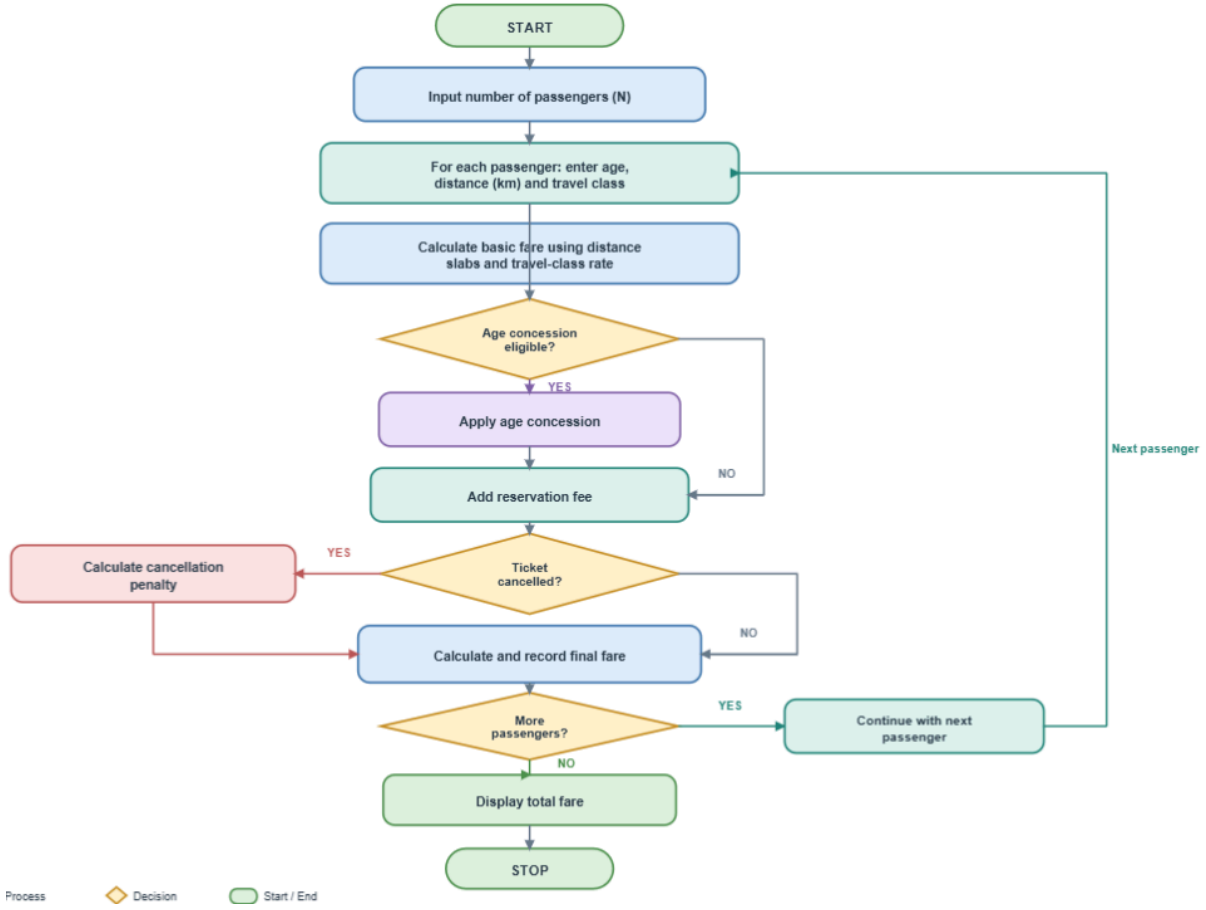
3. Assumptions

1. The user enters valid passenger details.
2. Distance is entered in kilometres.
3. Each travel class has a predefined fare rate.
4. Age-based concessions are applied according to the given rules.
5. Reservation fees and cancellation penalties are predefined.

4. Algorithm

1. Start.
2. Read the number of passengers.
3. Read each passenger's age, travel distance, and travel class.
4. Calculate the basic fare using distance slabs.
5. Apply the travel class fare.
6. Check the passenger's age and apply the eligible concession.
7. Add reservation fees.
8. If the ticket is cancelled, calculate the cancellation penalty.
9. Calculate the final fare for each passenger.
10. Display individual fares and the total amount.
11. Stop.

5. Flowchart



6. Pseudocode

None

```
START

INPUT number_of_passengers

FOR each passenger
    INPUT age
    INPUT distance
    INPUT travel_class

    CALCULATE basic_fare
    CALCULATE class_fare

    IF age satisfies concession rules THEN
        APPLY concession
    END IF
```

```
    ADD reservation_fee

    INPUT cancellation_status

    IF cancellation_status == "Yes" THEN
        CALCULATE cancellation_penalty
    END IF

    CALCULATE final_fare
    DISPLAY final_fare
END FOR

DISPLAY total_fare

STOP
```

8. Test Case and Expected Output

Input:

```
None
Enter number of passengers: 1

Passenger 1
Enter age: 65
Enter distance in km: 100
Select travel class: 1
Cancel ticket? (yes/no): no
```

Calculation:

```
None
Basic fare = 100 * 2 = 200
Senior concession = 20%
Fare after concession = 160
Reservation fee = 50
Final fare = 210
```

Expected Output:

```
None
Final fare: 210
Total fare: 210
```

9. Conclusion

The program calculates railway fares for multiple passengers using distance slabs, travel classes, age-based concessions, reservation fees, and cancellation penalties.